

Machine Learning

Unit-II (B): Multiclass Classification using SoftMax

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What is Softmax Function

- The Softmax function is a mathematical function that **converts a vector of real numbers into a probability distribution**.
- Each element in the output is between 0 and 1, and the **sum of all elements equals 1**.
- This property makes it perfect for classification tasks, where we want to know the probability that a given input belongs to a certain class.

The Formula for the Softmax Function

$$\text{Softmax}(z)_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Where:

- z_i represents the i^{th} raw score (also known as logits) of the model.
- K is the number of possible classes.
- The output of the function is the probability of the input belonging to each class.

How the Softmax Classifier Works?

- In a Softmax Classifier, the neural network outputs a set of raw scores for each class.
- These raw scores, also called **logits**, are then passed through the Softmax function, converting them into probabilities.
- The **class with the highest probability** is chosen as the model's prediction.
- For example,
 - If you're classifying an image of an animal into one of three categories (cat, dog, rabbit), the neural network might output raw scores like [2.1, 1.0, 0.5].
 - After applying the Softmax function, these scores might become [0.7, 0.2, 0.1], meaning there's a 70% chance the image is a cat, 20% chance it's a dog, and 10% chance it's a rabbit.

Example

A university wants to predict a student's preferred specialization based on their aptitude score x .

There are three specializations:

Class 1: Data Science (DS)

Class 2: Cyber Security (CS)

Class 3: Cloud Computing (CC)

Example

A multinomial logistic regression model (Softmax model) has learned the following parameters:

$$z_1 = 0.5x + 0.2$$

$$z_2 = 0.3x + 0.1$$

$$z_3 = -0.4x + 0.5$$

For a student with aptitude score $x = 2$:

1. Compute z_1, z_2, z_3
 2. Compute the SoftMax probabilities
 3. Identify the predicted specialization
- (Use values up to 3 decimal places)

Step 1: Compute Linear Scores

Given $x = 2$

$$z_1 = 0.5(2) + 0.2 = 1 + 0.2 = 1.2$$

$$z_2 = 0.3(2) + 0.1 = 0.6 + 0.1 = 0.7$$

$$z_3 = -0.4(2) + 0.5 = -0.8 + 0.5 = -0.3$$

Step 2: Compute Exponentials

- $e^{1.2} = 3.320$
- $e^{0.7} = 2.014$
- $e^{-0.3} = 0.741$

Step 3: Compute Denominator

$$\text{Denominator} = 3.320 + 2.014 + 0.741 = 6.075$$

Step 4: Compute Probabilities (Softmax)

- $P_1 = \frac{3.320}{6.075} = 0.547$

- $P_2 = \frac{2.014}{6.075} = 0.332$

- $P_3 = \frac{0.741}{6.075} = 0.122$

Step 5: Final Prediction

- Highest probability = **0.547**
- Predicted Specialization: Data Science (Class 1)

Question

A company wants to predict the type of job role suitable for a candidate based on:

x_1 : Technical Skill Score

x_2 : Communication Score

There are three job roles:

1. Class 1: Developer (D)
2. Class 2: Analyst (A)
3. Class 3: Manager (M)

Question

A trained multinomial logistic regression model has the following parameters:

Class 1 (Developer): $z_1 = 0.6x_1 + 0.2x_2 + 0.5$

Class 2 (Analyst): $z_2 = 0.3x_1 + 0.4x_2 + 0.2$

Class 3 (Manager): $z_3 = -0.2x_1 + 0.5x_2 + 0.1$

For a candidate with:

$$x_1 = 2, x_2 = 3$$

1. Compute z_1, z_2, z_3
2. Compute the Softmax probabilities
3. Identify the predicted job role

(Use values up to 3 decimal places.)

Step 1: Compute Linear Scores

Class 1

$$z_1 = 0.6(2) + 0.2(3) + 0.5$$

$$z_1 = 1.2 + 0.6 + 0.5 = 2.3$$

Class 2

$$z_2 = 0.3(2) + 0.4(3) + 0.2$$

$$z_2 = 0.6 + 1.2 + 0.2 = 2.0$$

Class 3

$$z_3 = -0.2(2) + 0.5(3) + 0.1$$

$$z_3 = -0.4 + 1.5 + 0.1 = 1.2$$

Step 2: Compute Exponentials

- $e^{2.3} = 9.974$
- $e^{2.0} = 7.389$
- $e^{1.2} = 3.320$

Step 3: Compute Denominator

$$\text{Denominator} = 9.974 + 7.389 + 3.320 = 20.683$$

Step 4: Compute Softmax Probabilities

- $P_1 = \frac{9.974}{20.683} = 0.482$

- $P_2 = \frac{7.389}{20.683} = 0.357$

- $P_3 = \frac{3.320}{20.683} = 0.161$

Step 5: Final Prediction

- Highest probability = 0.482
- Predicted Job Role: Developer

Softmax vs Sigmoid

Criteria	Softmax	Sigmoid
Purpose	Multi-class classification	Binary classification
Mathematical Expression	$\text{Softmax}(z)_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$	$\sigma(z) = \frac{1}{1 + e^{-z}}$
Output	A vector of probabilities for each class	A single probability for the positive class

Softmax vs Sigmoid

Criteria	Softmax	Sigmoid
Interpretation	Probabilities sum to 1, and the class with the highest probability is chosen	Output > 0.5 indicates the positive class, else negative
Use Case	Multi-class tasks (e.g., image classification with more than two categories)	Binary tasks (e.g., spam classification)
Handling Multiple Classes	Handles multiple classes in a mutually exclusive manner	Multiple Sigmoid functions needed, probabilities don't sum to 1
Loss Function	Categorical Cross-Entropy	Binary Cross-Entropy

Note: Cross-entropy is equivalent to Negative Log-Likelihood (NLL)

Loss Function: Cross-Entropy

- The Softmax Classifier is often paired with the Cross-Entropy Loss function, which is used to measure the error between the predicted probabilities and the true labels.
- Cross-entropy is defined as:

$$\text{Loss} = - \sum_{i=1}^K y_i \log(\hat{y}_i)$$

- Where:
- y_i is the true label (1 for the correct class and 0 for the others).
- $\hat{y}_i$ is the predicted probability for the i^{th} class.
- The Softmax function combined with the cross-entropy loss allows the model to penalize incorrect predictions while ensuring that the total probability sums to 1.

Question 1

A university is classifying students into three categories based on performance:

- Class 1: Distinction
 - Class 2: First Class
 - Class 3: Second Class
- For a particular student, the Softmax model outputs the following probabilities:

$$\hat{y} = [0.7, 0.2, 0.1]$$

- The true class is First Class.

Tasks:

1. Write the one-hot encoded true label.
2. Compute the Cross-Entropy Loss.

(Use natural logarithm and answer up to 3 decimal places.)

Solution 1

Predicted probabilities: $\hat{y} = [0.7, 0.2, 0.1]$ and True class = First Class (Class 2)

Step 1: One-hot encoding

$$y = [0, 1, 0]$$

Step 2: Cross-Entropy Loss

$$L = -\sum y_i \log(\hat{y}_i)$$

Only correct class contributes:

$$L = -\log(0.2) = -(-1.609) = 1.609$$

Final Answer:

$$L = 1.609$$

Question 2

A medical diagnosis system classifies diseases into:

- Class 1: Flu
 - Class 2: Dengue
 - Class 3: Malaria
- For a patient, the model predicts:

$$\hat{y} = [0.1, 0.8, 0.1]$$

- But the actual disease is Malaria.

Tasks:

1. Compute the Cross-Entropy Loss.
2. Explain why the loss value is large or small.
3. What would happen to the loss if the model predicted 0.9 for the correct class?

Solution 2

$\hat{y} = [0.1, 0.8, 0.1]$ and True class = Malaria (Class 3)

Step 1: One-hot encoding

$$y = [0, 0, 1]$$

Step 2: Loss

$$L = -\log(0.1) = 2.303$$

Interpretation:

- Loss is large because the model gave very low probability (0.1) to the correct class.

If model predicted 0.9:

$$L = -\log(0.9) = 0.105$$

- Much smaller $\rightarrow$ better prediction.

Question 3

An e-commerce platform classifies products into:

- Electronics
- Clothing
- Groceries

Two models give the following predictions for a product that actually belongs to Clothing:

Model A: $[0.2, 0.6, 0.2]$

Model B: $[0.4, 0.5, 0.1]$

Tasks:

1. Compute the Cross-Entropy Loss for both models.
2. Which model performs better? Justify numerically.

Solution 3

True class = Clothing (Class 2)

Model A:

$$[0.2, 0.6, 0.2]$$

$$L_A = -\log(0.6) = 0.511$$

Model B:

$$[0.4, 0.5, 0.1]$$

$$L_B = -\log(0.5) = 0.693$$

Comparison: $L_A < L_B$

Model A performs better.

Question 4

A Softmax classifier predicts the following probabilities for a 3-class problem:

$$\hat{y} = [0.5, 0.3, 0.2]$$

The true label is Class 1:

$$y = [1, 0, 0]$$

Tasks:

1. Compute the Cross-Entropy Loss.
2. Using the result $\frac{\partial L}{\partial z_i} = \hat{y}_i - y_i$, compute the gradient vector.
3. Interpret the sign of each gradient component.

Solution 4

$\hat{y} = [0.5, 0.3, 0.2]$ and True label is $y = [1, 0, 0]$

Step 1: Loss

$$L = -\log(0.5) = 0.693$$

Step 2: Gradient

$$\begin{aligned}\frac{\partial L}{\partial z_i} &= \hat{y}_i - y_i \\ &= [0.5 - 1, 0.3 - 0, 0.2 - 0] = [-0.5, 0.3, 0.2]\end{aligned}$$

Step 3: Interpretation

Negative gradient for correct class $\rightarrow$ increase its score

Positive gradient for wrong classes $\rightarrow$ decrease their scores

Question 5

A spam detection system predicts the probability of an email being spam as:

$$\hat{y} = 0.8$$

The true label is not spam (0).

Tasks:

1. Write the Binary Cross-Entropy formula.
2. Compute the loss.
3. Explain why the loss is high or low.

Solution 5

$$\hat{y} = 0.8 \text{ and True label} = 0$$

Step 1: Formula

$$L = -[y\log(\hat{y}) + (1 - y)\log(1 - \hat{y})]$$

Step 2: Substitute

$$L = -[0.\log(0.8) + 1.\log(0.2)] = -\log(0.2) = 1.609$$

Step 3: Interpretation

- Loss is high because model confidently predicted spam (0.8), but actual label is not spam.

Question 6

A 3-class Softmax classifier has the following predicted probabilities for one training example:

$$\hat{y} = [0.6, 0.3, 0.1]$$

The true class is Class 1.

The L2-regularized loss function is:

$$J = - \sum_{i=1}^K y_i \log(\hat{y}_i) + \frac{\lambda}{2m} \sum w^2$$

Given:

- $\lambda = 0.5$
- $m = 1$
- $\sum w^2 = 4$

Question 6

Tasks:

1. Compute the Cross-Entropy part.
2. Compute the regularization term.
3. Compute total loss.

Solution 6

$\hat{y} = [0.6, 0.3, 0.1]$ and True class = Class 1

Step 1: Cross-Entropy

$$L_{CE} = -\log(0.6) = 0.511$$

Step 2: Regularization Term

$$\frac{\lambda}{2m} \sum w^2 = \frac{0.5}{2(1)} \times 4 = \frac{0.5}{2} \times 4 = 0.25 \times 4 = 1$$

Step 3: Total Loss

$$J = 0.511 + 1 = 1.511$$

Applications of Softmax Classifier

1. **Image Classification:** It is used in Convolutional Neural Networks (CNNs) to classify images into multiple categories, such as identifying objects in an image.
2. **Natural Language Processing (NLP):** In models like LSTMs or Transformers, Softmax is used to predict the next word in a sequence or classify sentences.
3. **Speech Recognition:** Softmax is employed to convert raw audio data into probabilities for recognizing spoken words.

Advantages of the Softmax Classifier

1. **Multi-class Predictions:** It efficiently handles classification problems with more than two classes, making it suitable for real-world tasks like object detection, language processing, and more.
2. **Probabilistic Interpretation:** The output probabilities are easy to interpret, making it ideal for applications where confidence in the prediction is important.

Limitations of the Softmax Classifier

1. **Not Ideal for Binary Classification:** For problems with only two classes, the Sigmoid function and binary cross-entropy loss are more efficient.
2. **Sensitive to Outliers:** Since the Softmax function exponentiates its inputs, large values in the logits can disproportionately affect the output probabilities, leading to potential issues with outliers.

Thank You!